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FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don’t wait for motivation—show up with discipline. Because it’s not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

CONTENTS

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. ? + 37.5% of 960 = $\sqrt{5184} \times 16 - 4^3$

A. 728

B. 716

C. 742

D. 756

E. None of these

2. 68.75% of 512 + $\frac{2}{13}$ of 845 - $\sqrt[3]{15625}$ = ? + 332

A. 115

B. 125

C. 130

D. 135

E. None of these

3. $\frac{14}{15}$ of 675 + $\frac{5}{9}$ of 414 - $\frac{7}{16}$ of 704 = ?

A. 548

B. 556

C. 552

D. 560

E. None of these

4. ?% of 1250 + 27.27% of 1210 = 87.5% of 960 + 40

A. 42

B. 46

C. 48

D. 44

E. None of these

5. $(\sqrt{4096} + \sqrt{3136}) \times 9$ - 32.5% of 800 = ? × 20

A. 39

B. 43

C. 45

D. 47

E. None of these

6. $4\frac{5}{6} + 7\frac{1}{4} - 3\frac{1}{3} + 2\frac{7}{12}$ = ?

A. $10\frac{2}{3}$

B. $11\frac{5}{6}$

C. $11\frac{1}{3}$

D. $12\frac{1}{6}$

E. None of these

7. $\frac{5}{8}$ of ? - 56.25% of 320 = 145

A. 510

B. 520

C. 530

D. 540

E. None of these

8. $\sqrt[3]{110592}$ + 41.66% of 864 + $\frac{3}{10}$ of 750 = ? × 7 + 24

A. 83

B. 85

C. 89

D. 87

E. None of these

9. 72.72% of 990 - 31.25% of 896 + $\sqrt{7569}$ = 5 × ? + 2

A. 101

B. 105

C. 107

D. 109

E. None of these

10. [(26 × 18) + (32 × 14) - (24 × 11)] ÷ 4 = ? - 65

A. 228

B. 218

C. 224

D. 232

E. None of these

11. $?^2$ + 18.75% of 1536 = 75% of 864 + 9^2

A. 17

B. 19

C. 23

D. 21

E. None of these

12. 65% of 960 + 22.22% of 729 - ? = 68.75% of 640

A. 346

B. 326

C. 336

D. 356

E. None of these

13. ($\frac{5}{12}$ of 960 + $\frac{7}{15}$ of 480) ÷ ? = $\sqrt[3]{17576}$

A. 22

B. 24

C. 26

D. 28

E. None of these

14. 58.33% of 1152 + 44.44% of 540 - 6.25% of 1344 = ?

A. 824

B. 828

C. 832

D. 836

E. None of these

15. $\sqrt{?}$ - 24 × 7 = 12.5% of 800 + 2^5

A. 84100

B. 90000

C. 96100

D. 102400

E. None of these

16. 300% of 48 + 250% of 64 - 125% of 96 = ? × 8 + 24

A. 18

B. 22

C. 20

D. 24

E. None of these

17. (18.5 × 16) + (22.5 × 12) - (14.25 × 8) = ? × 4

A. 111

B. 109

C. 115

D. 113

E. None of these

18. $\frac{9}{14}$ of 1176 + $\frac{5}{11}$ of 726 - $\frac{7}{13}$ of 845 = ? + 210

A. 421

B. 411

C. 431

D. 441

E. None of these

19. $\frac{11}{16}$ of 704 + ? ÷ 8 = 83.33% of 720 + $\sqrt[3]{19683}$

A. 1104

B. 1124

C. 1144

D. 1164

E. None of these

20. 40% of (36^2 - 21^2) + 30% of (42^2 - 32^2) = ?

A. 554

B. 564

C. 574

D. 544

E. None of these

Answers:

1. A
2. B
3. C
4. D
5. E
6. C
7. B
8. D
9. B
10. A
11. D
12. A
13. B
14. B
15. B
16. C
17. D
18. A
19. C
20. B

Solutions:

1. ? + 37.5% of 960 = √5184 × 16 - 4^3

37.5% = $\frac{3}{8}$, √5184 = 72 and 4^3 = 64

37.5% of 960 = 360

So, ? + 360 = 72 × 16 - 64

? + 360 = 1152 - 64

? + 360 = 1088

? = 1088 - 360 = 728

2. 68.75% of 512 + $\frac{2}{13}$ of 845 - $\sqrt[3]{15625}$ = ? + 332

68.75% = $\frac{11}{16}$ and $\sqrt[3]{15625}$ = 25

$\frac{11}{16}$ of 512 = 352 and $\frac{2}{13}$ of 845 = 130

So, 352 + 130 - 25 = ? + 332

457 = ? + 332

? = 457 - 332 = 125

3. $\frac{14}{15}$ of 675 + $\frac{5}{9}$ of 414 - $\frac{7}{16}$ of 704 = ?

$\frac{14}{15}$ of 675 = 630, $\frac{5}{9}$ of 414 = 230 and $\frac{7}{16}$ of 704 = 308

So, 630 + 230 - 308 = ?

860 - 308 = ?

? = 552

4. ?% of 1250 + 27.27% of 1210 = 87.5% of 960 + 40

27.27% = $\frac{3}{11}$ and 87.5% = $\frac{7}{8}$

$\frac{3}{11}$ of 1210 = 330 and $\frac{7}{8}$ of 960 = 840

So, ?% of 1250 + 330 = 840 + 40

?% of 1250 = 880 - 330

?% of 1250 = 550

? = 550 × 100 ÷ 1250 = 44

5. (√4096 + √3136) × 9 - 32.5% of 800 = ? × 20

√4096 = 64, √3136 = 56 and 32.5% = $\frac{13}{40}$

$\frac{13}{40}$ of 800 = 260

So, (64 + 56) × 9 - 260 = ? × 20

120 × 9 - 260 = ? × 20

1080 - 260 = ? × 20

820 = ? × 20

? = 820 ÷ 20 = 41

6. $4\frac{5}{6} + 7\frac{1}{4} - 3\frac{1}{3} + 2\frac{7}{12}$ = ?

Taking LCM of 6, 4, 3 and 12 = 12

$4\frac{5}{6} = 4\frac{10}{12}$, $7\frac{1}{4} = 7\frac{3}{12}$, $3\frac{1}{3} = 3\frac{4}{12}$

So, 4 + 7 - 3 + 2 = 10

and $\frac{10}{12} + \frac{3}{12} - \frac{4}{12} + \frac{7}{12} = \frac{16}{12}$

$10 + \frac{16}{12} = 10 + 1\frac{1}{3}$

? = $11\frac{1}{3}$

7. $\frac{5}{8}$ of ? - 56.25% of 320 = 145

56.25% = $\frac{9}{16}$

$\frac{9}{16}$ of 320 = 180

So, $\frac{5}{8}$ of ? - 180 = 145

$\frac{5}{8}$ of ? = 145 + 180 = 325

? = 325 × 8 ÷ 5 = 520

8. $\sqrt[3]{110592}$ + 41.66% of 864 + $\frac{3}{10}$ of 750 = ? × 7 + 24

$\sqrt[3]{110592}$ = 48 and 41.66% = $\frac{5}{12}$

$\frac{5}{12}$ of 864 = 360 and $\frac{3}{10}$ of 750 = 225

So, 48 + 360 + 225 = ? × 7 + 24

633 = ? × 7 + 24

? × 7 = 633 - 24 = 609

? = 609 ÷ 7 = 87

9. 72.72% of 990 - 31.25% of 896 + √7569 = 5 × ? + 2

72.72% = $\frac{8}{11}$, 31.25% = $\frac{5}{16}$ and √7569 = 87

$\frac{8}{11}$ of 990 = 720 and $\frac{5}{16}$ of 896 = 280

So, 720 - 280 + 87 = 5 × ? + 2

527 = 5 × ? + 2

5 × ? = 525

? = 105

10. $[(26 \times 18) + (32 \times 14) - (24 \times 11)] \div 4 = ? - 65$

$26 \times 18 = 468$, $32 \times 14 = 448$ and $24 \times 11 = 264$

So, $(468 + 448 - 264) \div 4 = ? - 65$

$652 \div 4 = ? - 65$

$163 = ? - 65$

$? = 163 + 65 = 228$

11. $?^2 + 18.75\%$ of $1536 = 75\%$ of $864 + 9^2$

$18.75\% = \frac{3}{16}$, $75\% = \frac{3}{4}$ and $9^2 = 81$

$\frac{3}{16}$ of $1536 = 288$ and $\frac{3}{4}$ of $864 = 648$

So, $?^2 + 288 = 648 + 81$

$?^2 + 288 = 729$

$?^2 = 729 - 288 = 441$

$? = 21$

12. 65% of $960 + 22.22\%$ of $729 - ? = 68.75\%$ of 640

65% of $960 = 624$, $22.22\% = \frac{2}{9}$ and $68.75\% = \frac{11}{16}$

$\frac{2}{9}$ of $729 = 162$ and $\frac{11}{16}$ of $640 = 440$

So, $624 + 162 - ? = 440$

$786 - ? = 440$

$? = 786 - 440 = 346$

13. $(\frac{5}{12}$ of $960 + \frac{7}{15}$ of $480) \div ? = \sqrt[3]{17576}$

$\sqrt[3]{17576} = 26$

$\frac{5}{12}$ of $960 = 400$ and $\frac{7}{15}$ of $480 = 224$

So, $(400 + 224) \div ? = 26$

$624 \div ? = 26$

$? = 624 \div 26 = 24$

14. 58.33% of $1152 + 44.44\%$ of $540 - 6.25\%$ of $1344 = ?$

$58.33\% = \frac{7}{12}$, $44.44\% = \frac{4}{9}$ and $6.25\% = \frac{1}{16}$

$\frac{7}{12}$ of $1152 = 672$, $\frac{4}{9}$ of $540 = 240$ and $\frac{1}{16}$ of $1344 = 84$

So, $672 + 240 - 84 = ?$

$912 - 84 = ?$

$? = 828$

15. $\sqrt{?} - 24 \times 7 = 12.5\%$ of $800 + 2^5$

$12.5\% = \frac{1}{8}$ and $2^5 = 32$

$\frac{1}{8}$ of $800 = 100$ and $24 \times 7 = 168$

So, $\sqrt{?} - 168 = 100 + 32$

$\sqrt{?} - 168 = 132$

$\sqrt{?} = 132 + 168 = 300$

$? = 300^2 = 90000$

16. 300% of 48 + 250% of 64 - 125% of 96 = ? × 8 + 24

300% = 3, 250% = $\frac{5}{2}$ and 125% = $\frac{5}{4}$

300% of 48 = 144, 250% of 64 = 160 and 125% of 96 = 120

So, 144 + 160 - 120 = ? × 8 + 24

184 = ? × 8 + 24

? × 8 = 160

? = 20

17. (18.5 × 16) + (22.5 × 12) - (14.25 × 8) = ? × 4

18.5 × 16 = 296, 22.5 × 12 = 270 and 14.25 × 8 = 114

So, 296 + 270 - 114 = ? × 4

452 = ? × 4

? = 452 ÷ 4 = 113

18. $\frac{9}{14}$ of 1176 + $\frac{5}{11}$ of 726 - $\frac{7}{13}$ of 845 = ? + 210

$\frac{9}{14}$ of 1176 = 756, $\frac{5}{11}$ of 726 = 330 and $\frac{7}{13}$ of 845 = 455

So, 756 + 330 - 455 = ? + 210

631 = ? + 210

? = 631 - 210 = 421

19. $\frac{11}{16}$ of 704 + ? ÷ 8 = 83.33% of 720 + $\sqrt[3]{19683}$

$\frac{11}{16}$ of 704 = 484, 83.33% = $\frac{5}{6}$ and $\sqrt[3]{19683}$ = 27

$\frac{5}{6}$ of 720 = 600

So, 484 + ? ÷ 8 = 600 + 27

484 + ? ÷ 8 = 627

? ÷ 8 = 627 - 484 = 143

? = 143 × 8 = 1144

20. 40% of (36^2 - 21^2) + 30% of (42^2 - 32^2) = ?

36^2 - 21^2 = 1296 - 441 = 855

42^2 - 32^2 = 1764 - 1024 = 740

40% of 855 = 342 and 30% of 740 = 222

So, 342 + 222 = ?

? = 564

2. ARITHMETIC QUESTIONS

Q1: A trader sold a calculator at a profit of 28% and a speaker at a profit of 16%. The total profit earned from both items is Rs. 2520. If the ratio of the cost price of the calculator to that of the speaker is 5:7, then find the cost price of the calculator.
एक व्यापारी ने एक कैलकुलेटर को 28% लाभ पर और एक स्पीकर को 16% लाभ पर बेचा। दोनों वस्तुओं से अर्जित कुल लाभ Rs. 2520 है। यदि कैलकुलेटर और स्पीकर के क्रय मूल्य का अनुपात 5:7 है, तो कैलकुलेटर का क्रय मूल्य ज्ञात कीजिए।
(A) Rs. 4200
(B) Rs. 5000
(C) Rs. 5600
(D) Rs. 4800
(E) Rs. 6000

Q2: 18 men can complete a work in 24 days. If 4 women can do as much work as 3 men, then find how many days 36 women will take to complete the same work.
18 पुरुष किसी कार्य को 24 दिनों में पूरा कर सकते हैं। यदि 4 महिलाएँ उतना ही कार्य कर सकती हैं जितना 3 पुरुष करते हैं, तो 36 महिलाएँ उसी कार्य को कितने दिनों में पूरा करेंगी?
(A) 12 days
(B) 18 days
(C) 20 days
(D) 16 days
(E) 24 days

Q3: An investor invested Rs. P in Scheme A, which offers compound interest at 10% per annum for 2 years. The amount received from Scheme A was invested in Scheme B, which offers simple interest at 15% per annum for 4 years. If the interest received from Scheme B is Rs. 3630, find the value of P.
एक निवेशक ने योजना A में Rs. P निवेश किए, जिसमें 2 वर्षों के लिए 10% वार्षिक चक्रवृद्धि ब्याज मिलता है। योजना A से प्राप्त राशि को योजना B में निवेश किया गया, जिसमें 4 वर्षों के लिए 15% वार्षिक साधारण ब्याज मिलता है। यदि योजना B से प्राप्त ब्याज Rs. 3630 है, तो P का मान ज्ञात कीजिए।
(A) Rs. 5000

(B) Rs. 5500
(C) Rs. 6000
(D) Rs. 4500
(E) Rs. 6500

Q4: Pipe X alone fills 2/3 of a tank in 8 hours. After that, Pipe Y alone fills the remaining tank in 8 hours. Find the time taken by both pipes together to fill the whole tank when opened simultaneously.
पाइप X अकेले एक टंकी का 2/3 भाग 8 घंटों में भरता है। इसके बाद पाइप Y अकेले शेष टंकी को 8 घंटों में भरता है। यदि दोनों पाइप एक साथ खोले जाएँ, तो पूरी टंकी कितने समय में भरेगी?
(A) 6 hours
(B) 10 hours
(C) 12 hours
(D) 9 hours
(E) 8 hours

Q5: A shopkeeper mixes 6 kg of lentils costing Rs. 20 per kg with 4 kg of lentils costing Rs. 15 per kg. At what price per kg should he sell the mixture to make a profit of 100/3%?
एक दुकानदार Rs. 20 प्रति किग्रा वाली 6 किग्रा दाल को Rs. 15 प्रति किग्रा वाली 4 किग्रा दाल के साथ मिलाता है। 100/3% लाभ कमाने के लिए उसे मिश्रण को प्रति किग्रा किस मूल्य पर बेचना चाहिए?
(A) Rs. 18
(B) Rs. 21
(C) Rs. 24
(D) Rs. 27
(E) Rs. 30

Q6: Two investors jointly invested Rs 6000 in a business. The first investor kept his money for 12 months, while the second investor kept his money for 18 months. If the total profit was Rs 5800 and the profit share of the first investor was Rs 2800, find the investment of the second investor.
दो निवेशकों ने मिलकर एक व्यवसाय में रु. 6000 निवेश किए। पहले निवेशक ने अपनी राशि 12 महीनों के लिए रखी, जबकि दूसरे निवेशक ने अपनी राशि 18 महीनों के लिए रखी। यदि कुल लाभ रु. 5800 था और पहले निवेशक का लाभांश रु. 2800 था, तो दूसरे निवेशक का निवेश ज्ञात कीजिए।

(A) Rs 2200
(B) Rs 2700
(C) Rs 3000
(D) Rs 2500
(E) Rs 2000

Q7: Three years ago, the average age of Umesh and Charu was 27 years. Five years hence, the ratio of their ages will be 3:4. Find the sum of the terms of the ratio of their present ages.
तीन वर्ष पहले उमेश और चारु की औसत आयु 27 वर्ष थी। पांच वर्ष बाद उनकी आयु का अनुपात 3:4 होगा। उनकी वर्तमान आयु के अनुपात के पदों का योग ज्ञात कीजिए।
(A) 10
(B) 12
(C) 14
(D) 16
(E) 18

Q8: A train running at a speed of 72 km/h completely crosses a platform in 30 seconds. If the length of the platform is 25% of the length of the train, find the length of the platform.
72 किमी/घंटा की गति से चल रही एक ट्रेन एक प्लेटफॉर्म को 30 सेकंड में पूरी तरह पार करती है। यदि प्लेटफॉर्म की लंबाई ट्रेन की लंबाई की 25% है, तो प्लेटफॉर्म की लंबाई ज्ञात कीजिए।
(A) 180 meters
(B) 150 meters
(C) 100 meters
(D) 200 meters
(E) 120 meters

Q9: A seller sold a table for Rs 900 at a loss of 25%. If he had sold the same table at a profit equal to 1/6 of its cost price, what would have been its selling price?
एक विक्रेता ने एक मेज को रु. 900 में 25% हानि पर बेचा। यदि वह उसी मेज को उसके क्रय मूल्य के 1/6 के बराबर लाभ पर बेचता, तो उसका विक्रय मूल्य क्या होता?
(A) Rs 1400
(B) Rs 1350
(C) Rs 1500
(D) Rs 1250

(E) Rs 1450

Q10: The length of a rectangular park is 40% more than its breadth. If the area of the park is 3500 square meters, find the perimeter of the park.
एक आयताकार पार्क की लंबाई उसकी चौड़ाई से 40% अधिक है। यदि पार्क का क्षेत्रफल 3500 वर्ग मीटर है, तो पार्क का परिमाप ज्ञात कीजिए।
(A) 220 meters
(B) 260 meters
(C) 240 meters
(D) 280 meters
(E) 200 meters

Q11: A person invests Rs. X in Scheme A, which offers simple interest at 15% per annum for 4 years. He receives Rs. 5400 as total interest from the scheme. If he invests the same amount X in a bank that offers compound interest at 10% per annum for 2 years, find the total amount received from the bank.
एक व्यक्ति योजना A में Rs. X निवेश करता है, जिस पर 4 वर्षों के लिए 15% वार्षिक साधारण ब्याज मिलता है। उसे योजना से कुल Rs. 5400 ब्याज मिलता है। यदि वह वही राशि X एक बैंक में 2 वर्षों के लिए 10% वार्षिक चक्रवृद्धि ब्याज पर निवेश करता है, तो बैंक से प्राप्त कुल राशि ज्ञात कीजिए।
(A) Rs. 10290
(B) Rs. 10680
(C) Rs. 10890
(D) Rs. 11250
(E) Rs. 11520

Q12: The present age ratio of Person M to Person N is 4 : 3, and the present age of Person O is 5/3 of the present age of Person N. If the difference between the present ages of Person O and Person M is 12 years, find the age of Person M after 5 years.
व्यक्ति M और व्यक्ति N की वर्तमान आयु का अनुपात 4 : 3 है, और व्यक्ति O की वर्तमान आयु व्यक्ति N की वर्तमान आयु की 5/3 है। यदि व्यक्ति O और व्यक्ति M की वर्तमान आयु के बीच अंतर 12 वर्ष है, तो 5 वर्ष बाद व्यक्ति M की आयु ज्ञात कीजिए।
(A) 49 years
(B) 47 years
(C) 51 years
(D) 55 years

(E) 53 years

Q13: Two trains, Train P and Train Q, are running in opposite directions and cross each other in 26 seconds. Train P and Train Q cross a pole in 14 seconds and 35 seconds respectively. Find the sum of the terms in the simplified ratio of the length of Train P to the length of Train Q.
दो ट्रेनें, ट्रेन P और ट्रेन Q, विपरीत दिशाओं में चल रही हैं और एक-दूसरे को 26 सेकंड में पार करती हैं। ट्रेन P और ट्रेन Q एक खंभे को क्रमशः 14 सेकंड और 35 सेकंड में पार करती हैं। ट्रेन P की लंबाई और ट्रेन Q की लंबाई के सरल अनुपात के पदों का योग ज्ञात कीजिए।
(A) 11
(B) 12
(C) 15
(D) 17
(E) 13

Q14: A car covers 648 km at a constant speed. When its speed is increased by 18 km/hr, it takes 20% less time to cover the same distance. Find the time taken by the car to cover the distance at its original speed.
एक कार 648 किमी की दूरी समान गति से तय करती है। जब इसकी गति 18 किमी/घंटा बढ़ा दी जाती है, तो उसी दूरी को तय करने में 20% कम समय लगता है। मूल गति से वही दूरी तय करने में कार को कितना समय लगेगा?
(A) 6 hours
(B) 8 hours
(C) 10 hours
(D) 9 hours
(E) 12 hours

Q15: The cost prices of two gadgets P and Q are Rs. X and Rs. (X + 400), respectively. A retailer allows a discount of 25% on each gadget. After the discount, he earns 50% profit on gadget P and 20% profit on gadget Q. If the sum of the marked prices of both gadgets is Rs. 2800, find X.
दो गैजेट P और Q के क्रय मूल्य क्रमशः Rs. X और Rs. (X + 400) हैं। एक रिटेलर दोनों गैजेट पर 25% की छूट देता है। छूट के बाद, उसे गैजेट P पर 50% लाभ और गैजेट Q पर 20% लाभ होता है। यदि दोनों गैजेट के अंकित मूल्यों का योग Rs. 2800 है, तो X ज्ञात कीजिए।
(A) Rs. 550
(B) Rs. 600
(C) Rs. 650

(D) Rs. 700
(E) Rs. 800

Q16: The ratio of the upstream speed to the downstream speed of a boat is 3 : 5. The boat covers 36 km upstream and 60 km downstream in a total of 8 hours. Find the speed of the stream as a percentage of the downstream speed.
एक नाव की धारा के विपरीत चाल और धारा के अनुकूल चाल का अनुपात 3 : 5 है। नाव 36 किमी धारा के विपरीत और 60 किमी धारा के अनुकूल कुल 8 घंटों में तय करती है। धारा की चाल, धारा के अनुकूल चाल का कितने प्रतिशत है?
(A) 15%
(B) 25%
(C) 20%
(D) 30%
(E) 10%

Q17: A box contains 120 cards numbered from 1 to 120. One card is drawn at random. If the probability of getting a non-perfect-square numbered card is p/q in its lowest form, find the value of p + q.
एक डिब्बे में 1 से 120 तक क्रमांकित 120 कार्ड हैं। एक कार्ड यादृच्छिक रूप से निकाला जाता है। यदि गैर-पूर्ण-वर्ग क्रमांकित कार्ड मिलने की प्रायिकता p/q के न्यूनतम रूप में है, तो p + q का मान ज्ञात कीजिए।
(A) 19
(B) 21
(C) 17
(D) 25
(E) 23

Q18: A person spends 20% of his income on groceries. From the remaining amount, he spends 25% on rent and 15% on travel, and saves the rest. If the difference between his savings and his grocery expense is Rs. 8400, find his income.
एक व्यक्ति अपनी आय का 20% किराने पर खर्च करता है। शेष राशि में से वह 25% किराए पर और 15% यात्रा पर खर्च करता है तथा बाकी राशि बचाता है। यदि उसकी बचत और किराने के खर्च के बीच का अंतर Rs. 8400 है, तो उसकी आय ज्ञात कीजिए।
(A) Rs. 30000
(B) Rs. 36000
(C) Rs. 42000

(D) Rs. 24000
(E) Rs. 48000

Q19: The ratio of the curved surface area to the total surface area of a cylinder is 3 : 4. If the volume of the cylinder is 3234 cubic cm, find the height of the cylinder. Take π = 22/7.
एक बेलन के वक्र पृष्ठीय क्षेत्रफल और कुल पृष्ठीय क्षेत्रफल का अनुपात 3 : 4 है। यदि बेलन का आयतन 3234 घन सेमी है, तो बेलन की ऊँचाई ज्ञात कीजिए। π = 22/7 लीजिए।
(A) 14 cm
(B) 28 cm
(C) 35 cm
(D) 21 cm
(E) 42 cm

Q20: Worker A alone can complete a piece of work in 9 days, and Worker B alone can complete the same work in 18 days. If they work on alternate days starting with Worker A, find the number of days required to complete the whole work.
मजदूर A अकेले एक कार्य को 9 दिनों में पूरा कर सकता है और मजदूर B अकेले उसी कार्य को 18 दिनों में पूरा कर सकता है। यदि वे मजदूर A से शुरू करके बारी-बारी से काम करते हैं, तो पूरा कार्य कितने दिनों में पूरा होगा?
(A) 10 days
(B) 12 days
(C) 14 days
(D) 15 days
(E) 16 days

Answers:

1. B
2. D
3. A
4. E
5. C
6. D
7. B
8. E
9. A
10. C
11. C
12. E
13. E
14. D

15.B
16.C
17.E
18.A
19.D
20.B

Solutions:

1. Option B

Let the cost prices of the calculator and speaker be 5x and 7x respectively.
Profit on calculator = 28% of 5x
Profit on calculator = 28 × 5x/100 = 140x/100
Profit on speaker = 16% of 7x
Profit on speaker = 16 × 7x/100 = 112x/100
Total profit = Rs. 2520
So:
140x/100 + 112x/100 = 2520
252x/100 = 2520
x = 2520 × 100/252
x = 1000
Cost price of calculator = 5x
Cost price of calculator = 5 × 1000 = Rs. 5000

2. Option D

18 men can complete the work in 24 days.
Total work = 18 × 24 = 432 man-days
Given, 4 women can do the work of 3 men.
So, 36 women can do the work of 27 men.
Now, 27 men equivalent workers will complete 432 man-days of work.
Required time = 432/27
Required time = 16 days

3. Option A

Interest received from Scheme B = Rs. 3630
Rate of simple interest in Scheme B = 15% per annum
Time = 4 years

Let the amount invested in Scheme B be A.
Simple interest = Principal × Rate × Time/100
3630 = A × 15 × 4/100
3630 = A × 60/100
A = 3630 × 100/60
A = Rs. 6050
So, amount received from Scheme A = Rs. 6050
Scheme A gives compound interest at 10% per annum for 2 years.
Amount from Scheme A = P × 110/100 × 110/100
6050 = P × 121/100
P = 6050 × 100/121
P = Rs. 5000

4. Option E

Pipe X fills 2/3 of the tank in 8 hours.
Work done by Pipe X in 1 hour = 2/3 × 1/8
Work done by Pipe X in 1 hour = 1/12
Pipe Y fills the remaining 1/3 of the tank in 8 hours.
Work done by Pipe Y in 1 hour = 1/3 × 1/8
Work done by Pipe Y in 1 hour = 1/24
Work done by both pipes together in 1 hour = 1/12 + 1/24
Work done by both pipes together in 1 hour = 2/24 + 1/24
Work done by both pipes together in 1 hour = 3/24 = 1/8
So, both pipes together can fill the whole tank in 8 hours.

5. Option C

Cost of 6 kg lentils at Rs. 20 per kg = 6 × 20 = Rs. 120
Cost of 4 kg lentils at Rs. 15 per kg = 4 × 15 = Rs. 60
Total cost price = 120 + 60 = Rs. 180
Total quantity = 6 + 4 = 10 kg
Cost price per kg = 180/10 = Rs. 18
Required profit = 100/3%
Selling price per kg = 18 × (100 + 100/3)/100
Selling price per kg = 18 × 400/300
Selling price per kg = Rs. 24

6. Option D

Total investment = Rs 6000.
Let the investment of the first investor be Rs x.
So, investment of the second investor = Rs (6000 - x).
Total profit = Rs 5800.
Profit share of the first investor = Rs 2800.
Profit share of the second investor = 5800 - 2800 = Rs 3000.
Profit ratio of first investor and second investor = 2800:3000.
Simplify the ratio:
2800:3000 = 14:15.
In partnership, profit ratio = capital × time ratio.
So, x × 12 : (6000 - x) × 18 = 14 : 15.
Now cross multiply:
15 × 12x = 14 × 18 × (6000 - x)
180x = 252 × (6000 - x)
180x = 1512000 - 252x
180x + 252x = 1512000
432x = 1512000
x = 1512000/432
x = 3500
Investment of the first investor = Rs 3500.
Investment of the second investor = 6000 - 3500 = Rs 2500.

7. Option B

Three years ago, the average age of Umesh and Charu was 27 years.
So, their total age three years ago = 27 × 2 = 54 years.
Present total age = 54 + 3 + 3 = 60 years.
Five years hence, their total age will be:
60 + 5 + 5 = 70 years.
Five years hence, their age ratio will be 3:4.
Total ratio parts = 3 + 4 = 7 parts.
So, 7 parts = 70 years.
1 part = 70/7 = 10 years.
Their ages five years hence are:
3 parts = 3 × 10 = 30 years.
4 parts = 4 × 10 = 40 years.
Their present ages are:
30 - 5 = 25 years.
40 - 5 = 35 years.

Present age ratio = 25:35.
Simplify the ratio:
25:35 = 5:7.
Sum of the terms of the ratio = 5 + 7 = 12.

8. Option E

Speed of the train = 72 km/h.
Convert km/h into m/s:
$72 \times 5/18 = 20$ m/s.
The train crosses the platform in 30 seconds.
Distance covered in 30 seconds = $20 \times 30 = 600$ meters.
When a train crosses a platform:
Distance covered = length of train + length of platform.
Let the length of the train be x meters.
Platform length = 25% of train length = x/4.
So, total distance = x + x/4.
$x + x/4 = 600$
$5x/4 = 600$
$x = 600 \times 4/5$
x = 480 meters.
Length of platform = x/4 = 480/4 = 120 meters.

9. Option A

Selling price of the table at 25% loss = Rs 900.
At 25% loss, selling price = 75% of cost price.
So, cost price × 75/100 = 900.
Cost price = 900 × 100/75
Cost price = Rs 1200.
Now profit is equal to 1/6 of the cost price.
Profit = 1200 × 1/6 = Rs 200.
New selling price = Cost price + Profit.
New selling price = 1200 + 200 = Rs 1400.

10. Option C

Length is 40% more than breadth.
So, length = 140% of breadth = 7/5 of breadth.
Use the ratio shortcut:

Breadth : Length = 5 : 7.
Let breadth = 5x and length = 7x.
Area of rectangle = length × breadth.
So, $7x \times 5x = 3500$.
$35x^2 = 3500$.
$x^2 = 3500/35$.
$x^2 = 100$.
$x = 10$.
Breadth = 5x = 5 × 10 = 50 meters.
Length = 7x = 7 × 10 = 70 meters.
Perimeter of rectangle = 2 × (Length + Breadth).
Perimeter = 2 × (70 + 50)
Perimeter = 2 × 120
Perimeter = 240 meters.

11. Option C

Simple interest = Principal × Rate × Time/100.
Here, principal = Rs. X, rate = 15% per annum and time = 4 years.
So,
X × 15 × 4/100 = 5400
60X/100 = 5400
3X/5 = 5400
X = 5400 × 5/3
X = 9000.
So, the principal amount is Rs. 9000.
Now, this amount is invested at 10% compound interest for 2 years.
Amount = Principal × 110/100 × 110/100.
Amount = 9000 × 110/100 × 110/100
Amount = Rs. 10890.

12. Option E

Present age ratio of Person M and Person N = 4 : 3.
Let the present age of Person M = 4x.
Let the present age of Person N = 3x.
Person O's present age = 5/3 of Person N's present age.
Person O's present age = 5/3 × 3x = 5x.
Difference between the ages of Person O and Person M = 12 years.

So,
$5x - 4x = 12$
$x = 12$.
Present age of Person M = $4x = 4 \times 12 = 48$ years.
Age of Person M after 5 years = $48 + 5 = 53$ years.

13. Option E

Let the speed of Train P be p m/sec.
Let the speed of Train Q be q m/sec.
Train P crosses a pole in 14 seconds.
So, length of Train P = 14p.
Train Q crosses a pole in 35 seconds.
So, length of Train Q = 35q.
Both trains cross each other in opposite directions in 26 seconds.
So,
Length of Train P + Length of Train Q = Relative speed × Time
$14p + 35q = (p + q) \times 26$
$14p + 35q = 26p + 26q$
$35q - 26q = 26p - 14p$
$9q = 12p$
$p/q = 9/12 = 3/4$.
Now, ratio of lengths of Train P and Train Q:
Train P : Train Q = 14p : 35q
Put p : q = 3 : 4.
Train P : Train Q = 14 × 3 : 35 × 4
Train P : Train Q = 42 : 140
Train P : Train Q = 3 : 10.
Sum of the terms = 3 + 10 = 13.

14. Option D

Let the original speed of the car be x km/hr.
When speed is increased by 18 km/hr, new speed = x + 18 km/hr.
Time is reduced by 20%.
So, new time = 80% of original time.
For the same distance, speed and time are inversely proportional.
Therefore, new speed = 100/80 of original speed.
New speed = $5x/4$.

So,
5x/4 = x + 18
5x = 4x + 72
x = 72.
Original speed = 72 km/hr.
Distance = 648 km.
Original time = Distance/Speed.
Original time = 648/72 = 9 hours.

15. Option B

Cost price of gadget P = Rs. X.
Cost price of gadget Q = Rs. (X + 400).
Discount on each gadget = 25%.
So, selling price = 75% of marked price.
For gadget P:
Profit = 50%.
Selling price of P = 150% of X = 150X/100 = 3X/2.
Since selling price is 75% of marked price:
Marked price of P = Selling price × 100/75.
Marked price of P = 3X/2 × 100/75 = 2X.
For gadget Q:
Profit = 20%.
Selling price of Q = 120% of (X + 400).
Selling price of Q = 120/100 × (X + 400) = 6(X + 400)/5.
Marked price of Q = Selling price × 100/75.
Marked price of Q = 6(X + 400)/5 × 100/75.
Marked price of Q = 8(X + 400)/5.
Sum of marked prices = Rs. 2800.
So,
2X + 8(X + 400)/5 = 2800
10X/5 + 8X/5 + 3200/5 = 2800
18X/5 + 640 = 2800
18X/5 = 2160
18X = 10800
X = 10800/18
X = 600.

16. Option C

Let the upstream speed be 3x and the downstream speed be 5x.
Time taken upstream = 36/3x = 12/x hours
Time taken downstream = 60/5x = 12/x hours
Total time = 8 hours
So, 12/x + 12/x = 8
24/x = 8
x = 24/8 = 3
Upstream speed = 3 × 3 = 9 km/hr
Downstream speed = 5 × 3 = 15 km/hr
Speed of stream = (Downstream speed − Upstream speed)/2
Speed of stream = (15 − 9)/2 = 6/2 = 3 km/hr
Required percentage = Speed of stream × 100/Downstream speed
Required percentage = 3 × 100/15 = 20%

17. Option E

Total cards = 120
Perfect square numbers from 1 to 120 are:
1, 4, 9, 16, 25, 36, 49, 64, 81, 100
Number of perfect square cards = 10
Number of non-perfect-square cards = 120 − 10 = 110
Probability of getting a non-perfect-square card = 110/120
Simplify:
110/120 = 11/12
So, p = 11 and q = 12
p + q = 11 + 12 = 23

18. Option A

Let the income be Rs. 100.
Amount spent on groceries = 20% of 100 = Rs. 20
Remaining income = 100 − 20 = Rs. 80
From the remaining amount, he spends 25% on rent and 15% on travel.
Total spending from remaining amount = 25% + 15% = 40%
So, savings from remaining amount = 100% − 40% = 60%
Savings = 60% of Rs. 80
Savings = 80 × 60/100 = Rs. 48
Difference between savings and grocery expense = 48 − 20 = Rs. 28
When the difference is Rs. 28, income = Rs. 100

When the difference is Rs. 8400, income = 8400 × 100/28
Income = Rs. 30000

19. Option D

Curved surface area of cylinder = $2\pi rh$
Total surface area of cylinder = $2\pi r(h + r)$
Given ratio of CSA to TSA = 3 : 4
So, $2\pi rh/[2\pi r(h + r)] = 3/4$
$h/(h + r) = 3/4$
Cross multiply:
$4h = 3(h + r)$
$4h = 3h + 3r$
$h = 3r$
Volume of cylinder = $\pi r^2 h$
Put $h = 3r$.
Volume = $\pi \times r \times r \times 3r$
$3234 = 3\pi r^3$
Using $\pi = 22/7$:
$3234 = 3 \times 22/7 \times r^3$
$3234 = 66r^3/7$
$r^3 = 3234 \times 7/66$
$r^3 = 343$
r = 7 cm
Height = 3r = 3 × 7 = 21 cm

20. Option B

Worker A can complete the work in 9 days.
So, A’s one-day work = 1/9
Worker B can complete the work in 18 days.
So, B’s one-day work = 1/18
They work on alternate days starting with A.
Work done in 2 days = A’s one-day work + B’s one-day work
Work done in 2 days = 1/9 + 1/18
LCM of 9 and 18 = 18
Work done in 2 days = 2/18 + 1/18 = 3/18 = 1/6
So, in 2 days, they complete 1/6 of the work.

To complete the full work, required time = $6 \times 2 = 12$ days

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $x^2 - 3\sqrt{6}x + 12 = 0$
II. $y^2 - 2\sqrt{6}y + 6 = 0$

2.) I. $4x^2 - 31x + 55 = 0$
II. $7y^2 - 10y - 8 = 0$

3.) I. $6x^2 - 11x - 10 = 0$
II. $2y^2 - 27y + 90 = 0$

4.) I. $x^2 - 2\sqrt{5}x + 5 = 0$
II. $y^2 - 4\sqrt{5}y + 20 = 0$

5.) I. $8x^2 - 58x + 90 = 0$
II. $12y^2 - 19y - 18 = 0$

6.) I. $x^2 - 4\sqrt{3}x + 9 = 0$
II. $y^2 - 4\sqrt{3}y + 12 = 0$

7.) I. $8x^2 - 14x - 30 = 0$
II. $2y^2 - 17y + 33 = 0$

8.) I. $3x^2 - 28x + 49 = 0$
II. $4y^2 - 39y + 90 = 0$

9.) I. $x^2 - 5\sqrt{3}x + 18 = 0$
II. $y^2 - 2\sqrt{3}y + 3 = 0$

10.) I. $5x^2 - 43x + 90 = 0$
II. $6y^2 - 15y - 9 = 0$

11.) I. $x^2 - 2\sqrt{2}x + 2 = 0$
II. $y^2 - 7\sqrt{2}y + 12 = 0$

12.) I. $12x^2 - 15x - 18 = 0$
II. $3y^2 - 25y + 50 = 0$

13.) I. $x^2 - 5\sqrt{2}x + 8 = 0$
II. $y^2 - 6\sqrt{2}y + 18 = 0$

14.) I. $6x^2 - 43x + 65 = 0$
II. $2y^2 - 15y + 27 = 0$

15.) I. $x^2 - 2\sqrt{6}x + 6 = 0$
II. $y^2 - 5\sqrt{6}y + 36 = 0$

16.) I. $5x^2 - 38x + 65 = 0$
II. $15y^2 - 34y - 13 = 0$

17.) I. $x^2 - 4\sqrt{7}x + 28 = 0$
II. $y^2 - 3\sqrt{7}y + 14 = 0$

18.) I. $5x^2 - 4x - 28 = 0$
II. $10y^2 - 73y + 126 = 0$

19.) I. $x^2 - 7\sqrt{5}x + 50 = 0$
II. $y^2 - 2\sqrt{5}y + 5 = 0$

20.) I. $x^2 - 2\sqrt{7}x + 7 = 0$
II. $y^2 - 5\sqrt{7}y + 28 = 0$

Answers:

1. D
2. A
3. B
4. B
5. D
6. C
7. E
8. C
9. A
10. A
11. E
12. B
13. C
14. C
15. B
16. D
17. D
18. E
19. A
20. E

Solutions:

(1) $x = \sqrt{6}, 2\sqrt{6}$
$y = \sqrt{6}, \sqrt{6}$

(2) $x = \frac{11}{4}, 5$
$y = -\frac{4}{7}, 2$

(3) $x = -\frac{2}{3}, \frac{5}{2}$
$y = 6, \frac{15}{2}$

(4) $x = \sqrt{5}, \sqrt{5}$
$y = 2\sqrt{5}, 2\sqrt{5}$

(5) $x = \frac{9}{4}, 5$
$y = -\frac{2}{3}, \frac{9}{4}$

(6) $x = \sqrt{3}, 3\sqrt{3}$
$y = 2\sqrt{3}, 2\sqrt{3}$

(7) $x = -\frac{5}{4}, 3$
$y = 3, \frac{11}{2}$

(8) $x = \frac{7}{3}, 7$
$y = \frac{15}{4}, 6$

(9) $x = 2\sqrt{3}, 3\sqrt{3}$
$y = \sqrt{3}, \sqrt{3}$

(10) $x = \frac{18}{5}, 5$
$y = -\frac{1}{2}, 3$

(11) $x = \sqrt{2}, \sqrt{2}$
$y = \sqrt{2}, 6\sqrt{2}$

(12) $x = -\frac{3}{4}, 2$
$y = \frac{10}{3}, 5$

(13) $x = \sqrt{2}, 4\sqrt{2}$
$y = 3\sqrt{2}, 3\sqrt{2}$

(14) $x = \frac{13}{6}, 5$
$y = 3, \frac{9}{2}$

(15) $x = \sqrt{6}, \sqrt{6}$
$y = 2\sqrt{6}, 3\sqrt{6}$

(16) $x = \frac{13}{5}, 5$
$y = -\frac{1}{3}, \frac{13}{5}$

(17) $x = 2\sqrt{7}, 2\sqrt{7}$
$y = \sqrt{7}, 2\sqrt{7}$

(18) $x = -2, \frac{14}{5}$
$y = \frac{14}{5}, \frac{9}{2}$

(19) $x = 2\sqrt{5}, 5\sqrt{5}$
$y = \sqrt{5}, \sqrt{5}$

(20) $x = \sqrt{7}, \sqrt{7}$
$y = \sqrt{7}, 4\sqrt{7}$

4. WRONG NUMBER SERIES

(1) 4820, 4838, 4868, 4904, 4958, 5030

(a)5030

(b)4868

(c)4838

(d)4958

(e) None of these

(2) 9200, 9176, 9143, 9094, 9042, 8970

(a)8970

(b)9176

(c)9042

(d)9094

(e) None of these

(3) 348, 711, 1450, 2949, 5958, 11991

(a)1450

(b)711

(c)5958

(d)11991

(e) None of these

(4) 12500, 6285, 2140, 580, 183, 105.5

(a)183

(b)6285

(c)580

(d)2140

(e) None of these

(5) 760, 640, 805, 610, 900, 550

(a)640

(b)805

(c)610

(d)550

(e) None of these

(6) 860.5, 856.25, 849.5, 839.25, 830.5, 804.25

(a)830.5

(b)849.5

(c)804.25

(d)856.25

(e) None of these

(7) 75, 209, 804, 3976, 23768, 166296

(a)804

(b)209

(c)3976

(d)23768

(e) None of these

(8) 1440, 1451, 1471, 1500, 1546, 1605

(a)1546

(b)1451

(c)1605

(d)1500

(e) None of these

(9) 2330, 2407, 2498, 2617, 2755, 2911

(a)2407

(b)2755

(c)2330

(d)2911

(e) None of these

(10) 850, 865, 910, 1045, 1455, 2665

(a)850

(b)865

(c)910

(d)2665

(e) None of these

(11) 5000, 4964, 4896, 4800, 4656, 4460

(a)4656

(b)5000

(c)4896

(d)4460

(e) None of these

(12) 425, 825, 1600, 3140, 6150, 12175

(a)3140

(b)825

(c)12175

(d)6150

(e) None of these

(13) 300.5, 307, 320, 342, 398, 502

(a)300.5

(b)502

(c)398

(d)342

(e) None of these

(14) 3200, 1700, 930, 525, 300.5, 171.25

(a)930

(b)300.5

(c)171.25

(d)3200

(e) None of these

(15) 55, 325, 1605, 6445, 19315, 38605

(a)55

(b)6445

(c)19315

(d)38605

(e) None of these

(16) 910, 942, 990, 1066, 1170, 1332

(a)942

(b)1332

(c)1170

(d)1066

(e) None of these

(17) 570, 594, 675, 870, 1242, 1890

(a)594

(b)870

(c)1242

(d)1890

(e) None of these

(18) 500, 550, 660, 858, 1199.2, 1801.8

(a)1199.2

(b)660

(c)500

(d)1801.8

(e) None of these

(19) 820, 828, 801, 865, 748, 956

(a)820

(b)801

(c)748

(d)956

(e) None of these

(20) 1510, 1555, 1625, 1712, 1832, 1985

(a)1510

(b)1555

(c)1712

(d)1985

(e) None of these

Answers:

(1)b

(2)d

(3)a

(4)c

(5)e

(6)a

(7)c

(8)d

(9)b

(10)e

(11)c

(12)a

(13)d

(14)b

(15)e

(16)d

(17)b

(18)a

(19)c

(20)e

Solutions:

(1) +18, +27, +39, +54, +72

(2) −24, −33, −44, −57, −72

(3) ×2+15, ×2+30, ×2+45, ×2+60, ×2+75

(4) ÷2+35, ÷3+45, ÷4+55, ÷5+65, ÷6+75

(5) Odd positions: +45, +90; even positions: −30, −60

(6) −4.25, −6.75, −10.25, −14.75, −20.25

(7) ×3−16, ×4−32, ×5−48, ×6−64, ×7−80

(8) $+(3^2+2)$, $+(4^2+4)$, $+(5^2+6)$, $+(6^2+8)$, $+(7^2+10)$

(9) +(7×11), +(7×13), +(7×17), +(7×19), +(7×23)

(10) +15, +45, +135, +405, +1215

(11) -6^2, -8^2, -10^2, -12^2, -14^2

(12) ×2−25, ×2−50, ×2−75, ×2−100, ×2−125

(13) +6.5, +13, +26, +52, +104

(14) ÷2+100, ÷2+80, ÷2+60, ÷2+40, ÷2+20

(15) ×6−5, ×5−10, ×4−15, ×3−20, ×2−25

(16) +32, +48, +72, +108, +162

(17) $+(2^3\times3)$, $+(3^3\times3)$, $+(4^3\times3)$, $+(5^3\times3)$, $+(6^3\times3)$

(18) ×1.1, ×1.2, ×1.3, ×1.4, ×1.5

(19) $+2^3$, -3^3, $+4^3$, -5^3, $+6^3$

(20) +(5×9), +(6×11), +(7×13), +(8×15), +(9×17)

5. MISSING NUMBER SERIES

(1) 18, 24, 39, ?, 112, 178

(a) 63
(b) 65
(c) 67
(d) 69
(e) None of these

(2) 6, 19, 68, ?, 1442, 8671

(a) 279
(b) 283
(c) 285
(d) 287
(e) None of these

(3) 8188, 4101, ?, 355, 84, 29

(a) 1368
(b) 1376
(c) 1384
(d) 1392
(e) None of these

(4) 45, 67, 118, ?, 333, 554

(a) 189
(b) 193
(c) 195
(d) 197
(e) None of these

(5) 125, 135, 165, ?, 363, 585

(a) 233
(b) 235
(c) 237
(d) 239
(e) None of these

(6) 72.5, 75, 82.5, ?, 122.5, 160

(a) 95.5
(b) 97.5
(c) 99.5
(d) 101.5
(e) None of these

(7) 960, 949, 916, ?, 740, 575

(a) 846
(b) 848
(c) 850
(d) 852
(e) None of these

(8) 14, 27, 48, ?, 130, 199

(a) 77
(b) 79
(c) 80
(d) 81
(e) None of these

(9) 2560, 640, 1280, ?, 960, 240

(a) 310
(b) 320
(c) 330
(d) 340
(e) None of these

(10) 36, 61, 97, ?, 210, 291

(a) 142
(b) 144
(c) 146
(d) 148
(e) None of these

(11) 2800, 2776, 2728, ?, 2440, 2056

(a) 2624
(b) 2628
(c) 2630
(d) 2632
(e) None of these

(12) 3, 12, 40, ?, 231, 448

(a) 105
(b) 107
(c) 109
(d) 111
(e) None of these

(13) 247, 269, 308, 376, ?, 609

(a) 465
(b) 467
(c) 469
(d) 471
(e) None of these

(14) 1820, 1800, 1755, ?, 1550, 1370

(a) 1665
(b) 1675
(c) 1685
(d) 1695
(e) None of these

(15) 58, 118, 242, ?, 1012, 2054

(a) 488
(b) 492
(c) 496
(d) 500
(e) None of these

(16) 720, 730, 760, 850, ?, 1930

(a) 1080
(b) 1100
(c) 1120
(d) 1140
(e) None of these

(17) 93, 103, 124, ?, 215, 293

(a) 154
(b) 160
(c) 166
(d) 172
(e) None of these

(18) 1218, 606, 298, ?, 62, 20

(a) 142
(b) 144
(c) 146
(d) 148
(e) None of these

(19) 42, 60, 114, ?, 762, 2220

(a) 268
(b) 272
(c) 274
(d) 276
(e) None of these

(20) 5600, 5583, 5549, ?, 5430, 5345

(a) 5492
(b) 5494
(c) 5498
(d) 5500
(e) None of these

Answers:

(1) c
(2) c

(3) b
(4) d
(5) a
(6) b
(7) c
(8) d
(9) b
(10) c
(11) d
(12) a
(13) d
(14) b
(15) c
(16) c
(17) b
(18) a
(19) d
(20) c

Solutions:

(1) $+2\times3, +3\times5, +4\times7, +5\times9, +6\times11$

(2) $\times2 + 7, \times3 + 11, \times4 + 13, \times5 + 17, \times6 + 19$

(3) $\div2 + 7, \div3 + 9, \div4 + 11, \div5 + 13, \div6 + 15$

(4) Sum of previous two numbers +6, +12, +18, +24

(5) $+2^3 + 2, +3^3 + 3, +4^3 + 4, +5^3 + 5, +6^3 + 6$

(6) $+2.5\times1, +2.5\times3, +2.5\times6, +2.5\times10, +2.5\times15$

(7) $-11\times1, -11\times3, -11\times6, -11\times10, -11\times15$

(8) +13, +21, +33, +49, +69
Second difference: +8, +12, +16, +20

(9) $\div4, \times2, \div4, \times3, \div4$

(10) $+5^2, +6^2, +7^2, +8^2, +9^2$

(11) −24, −48, −96, −192, −384

(12) $+2^3 + 1, +3^3 + 1, +4^3 + 1, +5^3 + 1, +6^3 + 1$

(13) $+2\times11, +3\times13, +4\times17, +5\times19, +6\times23$

(14) $-5\times2^2, -5\times3^2, -5\times4^2, -5\times5^2, -5\times6^2$

(15) $\times2 + 2, \times2 + 6, \times2 + 12, \times2 + 20, \times2 + 30$

(16) +10, +30, +90, +270, +810

(17) $+2\times5, +3\times7, +4\times9, +5\times11, +6\times13$

(18) $\div2 - 3, \div2 - 5, \div2 - 7, \div2 - 9, \div2 - 11$

(19) +18, +54, +162, +486, +1458

(20) −17, −34, −51, −68, −85

6. DATA INTERPRETATION

Set 1: The table shows the total number of lockers available in five different bank branches and the number of vacant lockers in these branches. Read the data carefully and answer the questions. Note: Total lockers available = Allotted lockers + Vacant lockers.
तालिका पाँच अलग-अलग बैंक शाखाओं में उपलब्ध कुल लॉकरों और खाली लॉकरों की संख्या दर्शाती है। आँकड़ों को ध्यानपूर्वक पढ़िए और प्रश्नों के उत्तर दीजिए। नोट: उपलब्ध कुल लॉकर = आवंटित लॉकर + खाली लॉकर।

Bank Branch / बैंक शाखा	Total lockers available / उपलब्ध कुल लॉकर	Vacant lockers / खाली लॉकर
Agra Branch / आगरा शाखा	826	238
Bhopal Branch / भोपाल शाखा	875	371
Cuttack Branch / कटक शाखा	1008	392
Raipur Branch / रायपुर शाखा	1134	350
Mysuru Branch / मैसूर शाखा	1148	420

Note: Total lockers available = Allotted lockers + Vacant lockers.
नोट: उपलब्ध कुल लॉकर = आवंटित लॉकर + खाली लॉकर।

Q1: Find the average number of allotted lockers in Bhopal Branch, Cuttack Branch and Mysuru Branch together.
भोपाल शाखा, कटक शाखा और मैसूर शाखा में मिलाकर आवंटित लॉकरों की औसत संख्या ज्ञात कीजिए।
(A) 602
(B) 630
(C) 616
(D) 644
(E) 588

Q2: In Pune Branch, the total number of available lockers is 25% more than that in Cuttack Branch, and the number of allotted lockers is 40% more than the vacant

lockers in Mysuru Branch. Find the total number of vacant lockers in Pune Branch.
पुणे शाखा में उपलब्ध कुल लॉकर, कटक शाखा की तुलना में 25% अधिक हैं और आवंटित लॉकरों की संख्या, मैसूर शाखा के खाली लॉकरों की तुलना में 40% अधिक है। पुणे शाखा में खाली लॉकरों की कुल संख्या ज्ञात कीजिए।

(A) 630
(B) 658
(C) 700
(D) 686
(E) 672

Q3: The annual rent of one allotted locker in Agra Branch is Rs. X and in Cuttack Branch is Rs. Y. If the total rent collected by Agra Branch is Rs. 142296 and this rent is 10% more than the rent collected by Cuttack Branch, then find X + Y.
आगरा शाखा में एक आवंटित लॉकर का वार्षिक किराया X रुपये और कटक शाखा में Y रुपये है। यदि आगरा शाखा द्वारा प्राप्त कुल किराया 142296 रुपये है और यह किराया कटक शाखा द्वारा प्राप्त किराए से 10% अधिक है, तो X + Y ज्ञात कीजिए।

(A) 452
(B) 440
(C) 464
(D) 476
(E) 488

Q4: The number of vacant lockers in Raipur Branch is what percentage of the total number of lockers available in Bhopal Branch?
रायपुर शाखा में खाली लॉकरों की संख्या, भोपाल शाखा में उपलब्ध कुल लॉकरों का कितने प्रतिशत है?

(A) 45%
(B) 40%
(C) 48%
(D) 52%
(E) 56%

Q5: Find by how many the allotted lockers in Agra Branch and Raipur Branch together are more or less than the allotted lockers in Mysuru Branch and Bhopal Branch together.
आगरा शाखा और रायपुर शाखा में मिलाकर आवंटित लॉकर, मैसूर शाखा और भोपाल शाखा में मिलाकर आवंटित लॉकरों से कितने अधिक या कम हैं?

(A) 84
(B) 112
(C) 126
(D) 140
(E) 154

Solutions:

Branch / शाखा	Total lockers / कुल लॉकर	Vacant lockers / खाली लॉकर	Allotted lockers / आवंटित लॉकर
Agra Branch / आगरा शाखा	826	238	588
Bhopal Branch / भोपाल शाखा	875	371	504
Cuttack Branch / कटक शाखा	1008	392	616
Raipur Branch / रायपुर शाखा	1134	350	784
Mysuru Branch / मैसूर शाखा	1148	420	728

Common values:
Agra Branch: Allotted lockers = 826 - 238 = 588
Bhopal Branch: Allotted lockers = 875 - 371 = 504
Cuttack Branch: Allotted lockers = 1008 - 392 = 616
Raipur Branch: Allotted lockers = 1134 - 350 = 784
Mysuru Branch: Allotted lockers = 1148 - 420 = 728

1. Option C
Allotted lockers in Bhopal, Cuttack and Mysuru = 504 + 616 + 728 = 1848
Average = 1848/3 = 616

2. Option E
Total lockers in Pune Branch = 125% of 1008 = 1260
Allotted lockers in Pune Branch = 140% of 420 = 588
Vacant lockers in Pune Branch = 1260 - 588 = 672

3. Option A
X = 142296/588 = 242
Rent collected by Cuttack Branch = 142296 × 10/11 = 129360
Y = 129360/616 = 210
X + Y = 242 + 210 = 452

4. Option B
Required percentage = 350/875 × 100
Required percentage = 40%

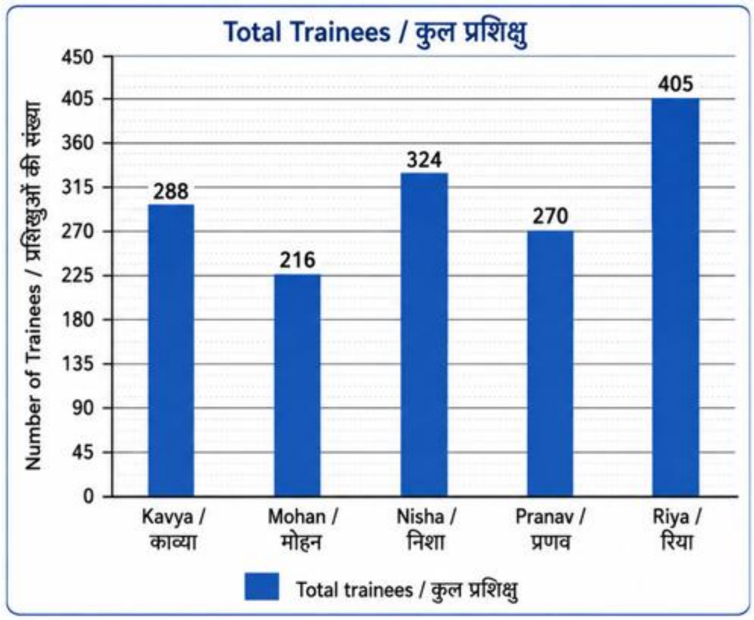
5. Option D
Allotted lockers in Agra and Raipur = 588 + 784 = 1372
Allotted lockers in Mysuru and Bhopal = 728 + 504 = 1232
Required difference = 1372 - 1232 = 140

Set 2: The bar graph shows the total number of trainees, boys and girls together, enrolled in five different skill-training centres. The table shows the percentage of girls out of the total trainees in these centres.

बार ग्राफ पाँच अलग-अलग कौशल-प्रशिक्षण केंद्रों में नामांकित कुल प्रशिक्षुओं, अर्थात लड़के और लड़कियाँ मिलाकर, की संख्या दिखाता है। तालिका इन केंद्रों में कुल प्रशिक्षुओं में से लड़कियों का प्रतिशत दिखाती है।

Note: Total trainees = boys + girls.

नोट: कुल प्रशिक्षु = लड़के + लड़कियाँ।



Percentage Table / प्रतिशत तालिका

Training Centre / प्रशिक्षण केंद्र	Percentage of girls / लड़कियों का प्रतिशत
Kavya / काव्या	$43\frac{3}{4}\%$
Mohan / मोहन	$33\frac{1}{3}\%$
Nisha / निशा	$44\frac{4}{9}\%$
Pranav / प्रणव	$66\frac{2}{3}\%$
Riya / रिया	$66\frac{2}{3}\%$

Q1: The average number of girls in Mohan and Nisha together is what percentage more or less than the total number of boys in Riya?

मोहन और निशा में लड़कियों की औसत संख्या, रिया में लड़कों की कुल संख्या से कितने प्रतिशत अधिक या कम है?

(A) 25% less
(B) 20% less
(C) 15% more
(D) 30% less
(E) 10% more

Q2: The total number of boys in Samar Centre is 15 more than the average number of girls in Riya and Pranav. If the total number of trainees in Samar Centre is $33\frac{1}{3}\%$ more than that in Kavya, then find the total number of girls in Samar Centre.

समर केंद्र में लड़कों की कुल संख्या, रिया और प्रणव में लड़कियों की औसत संख्या से 15 अधिक है। यदि समर केंद्र में कुल प्रशिक्षुओं की संख्या काव्या की तुलना में $33\frac{1}{3}\%$ अधिक

है, तो समर केंद्र में लड़कियों की कुल संख्या ज्ञात कीजिए।
(A) 126
(B) 171
(C) 135
(D) 153
(E) 144

Q3: If each boy and each girl attend 9 hours and 7 hours of training per day, respectively, in Nisha Centre for 6 days in a week, then find the difference between the total hours attended by boys and girls in Nisha Centre in that week.
यदि निशा केंद्र में प्रत्येक लड़का और प्रत्येक लड़की एक दिन में क्रमशः 9 घंटे और 7 घंटे प्रशिक्षण लेते हैं, और सप्ताह में 6 दिन प्रशिक्षण होता है, तो उस सप्ताह निशा केंद्र में लड़कों और लड़कियों द्वारा लिए गए कुल प्रशिक्षण घंटों के बीच का अंतर ज्ञात कीजिए।
(A) 3672
(B) 3888
(C) 3600
(D) 3744
(E) 3528

Q4: In Kavya Centre, $\frac{2}{9}$ of the total boys and $\frac{4}{7}$ of the total girls choose the AI module. The remaining trainees choose the Data module. Find the average number of boys and girls who choose the Data module in Kavya Centre.
काव्या केंद्र में कुल लड़कों के $\frac{2}{9}$ भाग और कुल लड़कियों के $\frac{4}{7}$ भाग ने AI मॉड्यूल चुना। शेष प्रशिक्षुओं ने Data मॉड्यूल चुना। काव्या केंद्र में Data मॉड्यूल चुनने वाले लड़कों और लड़कियों की औसत संख्या ज्ञात कीजिए।
(A) 72
(B) 84
(C) 96
(D) 90
(E) 108

Q5: Find the ratio of the total number of boys in Nisha, Mohan and Riya together to the total number of trainees in Kavya and Mohan together. If the ratio is p:q in simplest form, find p + q.
निशा, मोहन और रिया में मिलाकर लड़कों की कुल संख्या का, काव्या और मोहन में मिलाकर कुल प्रशिक्षुओं की संख्या से अनुपात ज्ञात कीजिए। यदि सरलतम रूप में अनुपात p:q है, तो p + q ज्ञात कीजिए।

(A) 101
(B) 113
(C) 107
(D) 119
(E) 115

Solutions:

Training Centre / प्रशिक्षण केंद्र	Total / कुल	Girls % / लड़कियाँ %	Girls / लड़कियाँ	Boys / लड़के
Kavya / काव्या	288	43 3/4%	126	162
Mohan / मोहन	216	33 1/3%	72	144
Nisha / निशा	324	44 4/9%	144	180
Pranav / प्रणव	270	66 2/3%	180	90
Riya / रिया	405	66 2/3%	270	135

1. Option B
Girls in Mohan = 216 × $\frac{1}{3}$ = 72
Girls in Nisha = 324 × $\frac{4}{9}$ = 144
Average girls = (72 + 144) ÷ 2 = 108
Girls in Riya = 405 × $\frac{2}{3}$ = 270
Boys in Riya = 405 - 270 = 135
Difference = 135 - 108 = 27
Required percentage = 27/135 × 100 = 20% less

2. Option E
Girls in Riya = 405 × $\frac{2}{3}$ = 270
Girls in Pranav = 270 × $\frac{2}{3}$ = 180
Average girls = (270 + 180) ÷ 2 = 225
Boys in Samar = 225 + 15 = 240
Total trainees in Samar = 288 × 4/3 = 384
Girls in Samar = 384 - 240 = 144

3. Option A
Girls in Nisha = 324 × $\frac{4}{9}$ = 144
Boys in Nisha = 324 - 144 = 180
Boys' weekly hours = 180 × 9 × 6 = 9720
Girls' weekly hours = 144 × 7 × 6 = 6048

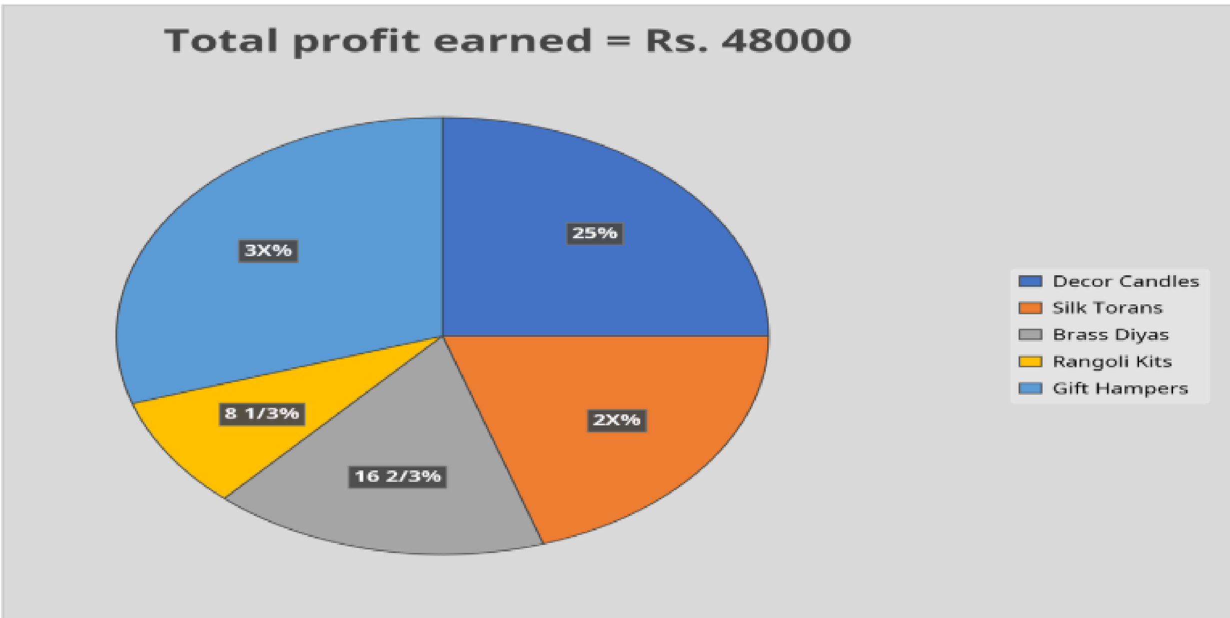
Difference = 9720 - 6048 = 3672

4. Option D
Girls in Kavya = 288 × $\frac{7}{16}$ = 126
Boys in Kavya = 288 - 126 = 162
Boys in AI module = 162 × $\frac{2}{9}$ = 36
Girls in AI module = 126 × $\frac{4}{7}$ = 72
Boys in Data module = 162 - 36 = 126
Girls in Data module = 126 - 72 = 54
Required average = (126 + 54) ÷ 2 = 90

5. Option C
Boys in Nisha = 324 - 144 = 180
Boys in Mohan = 216 - 72 = 144
Boys in Riya = 405 - 270 = 135
Total boys in Nisha, Mohan and Riya = 180 + 144 + 135 = 459
Total trainees in Kavya and Mohan = 288 + 216 = 504
Required ratio = 459:504 = 51:56
p + q = 51 + 56 = 107

Set 3: The pie chart shows the percentage distribution of profit earned by a company by selling five different decorative items on Monday. The total profit earned by selling all five items on Monday is Rs. 48,000.

पाई चार्ट एक कंपनी द्वारा सोमवार को पाँच अलग-अलग सजावटी वस्तुओं को बेचकर अर्जित लाभ का प्रतिशत वितरण दर्शाता है। सोमवार को सभी पाँच वस्तुओं को बेचकर अर्जित कुल लाभ Rs. 48,000 है।



Q1: Find the ratio of the profit earned on selling Gift Hampers to the profit earned on selling Silk Torans on Monday.
सोमवार को उपहार हैम्पर बेचकर अर्जित लाभ का रेशमी तोरण बेचकर अर्जित लाभ से अनुपात ज्ञात कीजिए।

(A) 4:3
(B) 5:4
(C) 7:5
(D) 3:2
(E) 2:1

Q2: The profit earned on selling Brass Diyas on Thursday is 75% more than the profit earned on selling Silk Torans on Monday. Find the difference between the profit earned on selling Rangoli Kits on Monday and the profit earned on selling Brass Diyas on Thursday.
गुरुवार को पीतल दीये बेचकर अर्जित लाभ, सोमवार को रेशमी तोरण बेचकर अर्जित लाभ से 75% अधिक है। सोमवार को रंगोली किट बेचकर अर्जित लाभ और गुरुवार को पीतल दीये बेचकर अर्जित लाभ के बीच का अंतर ज्ञात कीजिए।

(A) Rs. 12800
(B) Rs. 11800
(C) Rs. 13600
(D) Rs. 12400
(E) Rs. 13200

Q3: The profit on each Gift Hamper and each Silk Toran is Rs. 180 and Rs. 80, respectively. Find the ratio of the total number of Gift Hampers sold to the total number of Silk Torans sold on Monday.
प्रत्येक उपहार हैम्पर और प्रत्येक रेशमी तोरण पर लाभ क्रमशः Rs. 180 और Rs. 80 है। सोमवार को बेचे गए उपहार हैम्पर की कुल संख्या का बेचे गए रेशमी तोरणों की कुल संख्या से अनुपात ज्ञात कीजिए।
(A) 3:5
(B) 4:5
(C) 5:6
(D) 3:4
(E) 2:3

Q4: The profit earned on selling Gift Hampers on Tuesday to that on Monday is in the ratio 7:9. The profit earned on selling Gift Hampers on Tuesday is what percentage of the profit earned on selling Brass Diyas on Monday?
मंगलवार को उपहार हैम्पर बेचकर अर्जित लाभ और सोमवार को उपहार हैम्पर बेचकर अर्जित लाभ का अनुपात 7:9 है। मंगलवार को उपहार हैम्पर बेचकर अर्जित लाभ, सोमवार को पीतल दीये बेचकर अर्जित लाभ का कितना प्रतिशत है?
(A) 135%
(B) 140%
(C) 150%
(D) 125%
(E) 160%

Q5: Find the central angle, in degrees, for the profit earned on selling Silk Torans on Monday.
सोमवार को रेशमी तोरण बेचकर अर्जित लाभ के लिए केंद्रीय कोण, डिग्री में, ज्ञात कीजिए।
(A) 60
(B) 90
(C) 72
(D) 108

(E) 84

Solutions:

Item	Chart label	Solved percentage	Profit on Monday	Pie-chart value
Decor Candles	25%	25%	Rs. 12000	60
Silk Torans	2X%	20%	Rs. 9600	48
Brass Diyas	16 2/3%	16 2/3%	Rs. 8000	40
Rangoli Kits	8 1/3%	8 1/3%	Rs. 4000	20
Gift Hampers	3X%	30%	Rs. 14400	72
Total	**100%**	**100%**	**Rs. 48000**	**240**

Common Calculation:
25% + 16 2/3% + 8 1/3% + 2X% + 3X% = 100%
50% + 5X% = 100%
5X% = 50%
X = 10
Silk Torans = 2X% = 20%
Gift Hampers = 3X% = 30%
Total profit = Rs. 48000
Decor Candles profit = 25% of 48000 = Rs. 12000
Silk Torans profit = 20% of 48000 = Rs. 9600
Brass Diyas profit = 16 2/3% of 48000 = Rs. 8000
Rangoli Kits profit = 8 1/3% of 48000 = Rs. 4000
Gift Hampers profit = 30% of 48000 = Rs. 14400

1. Option D
Gift Hampers profit = Rs. 14400
Silk Torans profit = Rs. 9600
Required ratio = 14400:9600 = 3:2

2. Option A
Silk Torans profit on Monday = Rs. 9600
Brass Diyas profit on Thursday = 175% of 9600 = Rs. 16800
Rangoli Kits profit on Monday = Rs. 4000
Required difference = 16800 − 4000 = Rs. 12800

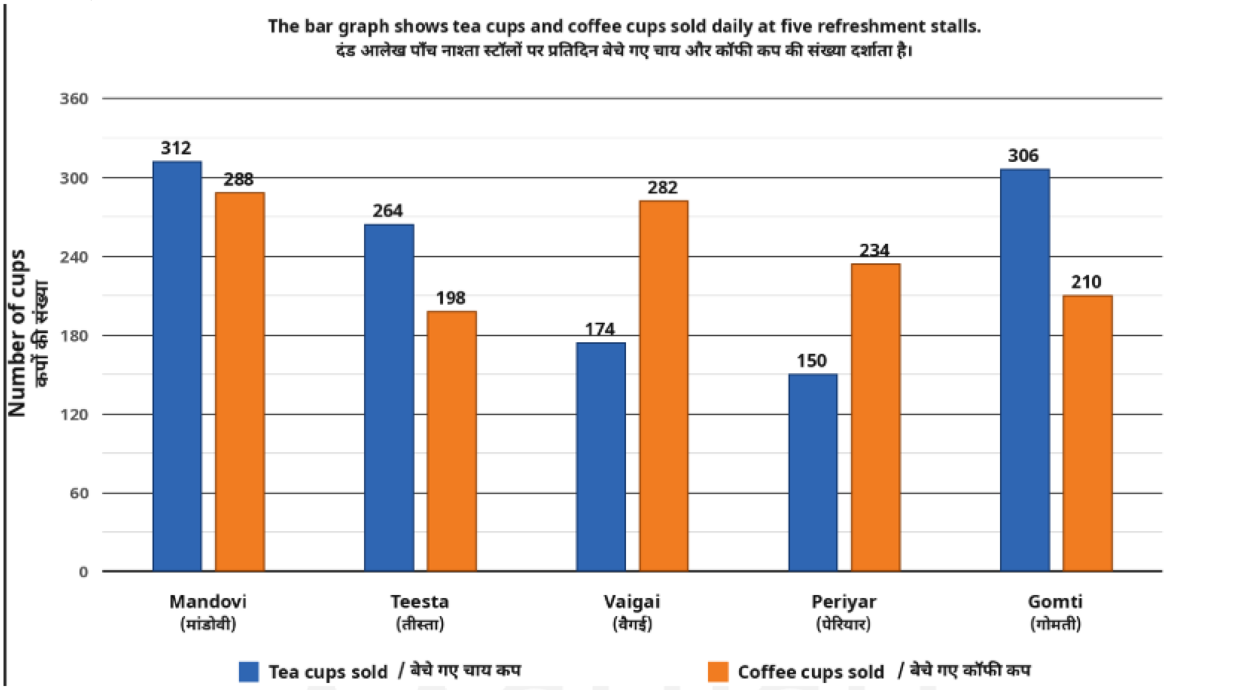
3. Option E
Gift Hampers profit = Rs. 14400
Silk Torans profit = Rs. 9600
Number of Gift Hampers sold = 14400/180 = 80
Number of Silk Torans sold = 9600/80 = 120
Required ratio = 80:120 = 2:3

4. Option B
Gift Hampers profit on Monday = Rs. 14400
Gift Hampers profit on Tuesday = 14400 × 7/9 = Rs. 11200
Brass Diyas profit on Monday = Rs. 8000
Required percentage = 11200/8000 × 100 = 140%

5. Option C
Silk Torans profit percentage = 20%
Central angle = 20% of 360
Central angle = 72 degrees

Set 4: The bar graph shows the total number of tea cups and coffee cups sold daily at five different refreshment stalls. Read the bar graph carefully and answer the following questions.

दंड आलेख पाँच अलग-अलग नाश्ता स्टॉलों पर प्रतिदिन बेचे गए चाय कप और कॉफी कप की कुल संख्या दर्शाता है। दंड आलेख को ध्यानपूर्वक पढ़ें और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: At a new refreshment stall Nayan, the number of tea cups sold is 1 ¼ times the number of coffee cups sold at Mandovi, and the number of coffee cups sold at Nayan is 37 ½% more than the number of tea cups sold at Teesta. Find how many more total cups are sold at Nayan than coffee cups sold at Periyar.

एक नए नाश्ता स्टॉल नयन पर बेचे गए चाय कप, मांडोवी में बेचे गए कॉफी कप का 1 ¼ गुना हैं, और नयन पर बेचे गए कॉफी कप, तीस्ता में बेचे गए चाय कप से 37 ½% अधिक हैं। ज्ञात कीजिए कि नयन पर बेचे गए कुल कप, पेरियार में बेचे गए कॉफी कप से कितने अधिक हैं।

(A) 459
(B) 471
(C) 483
(D) 489
(E) 501

Q2: Total coffee cups sold at Vaigai and Periyar together are approximately what percent more than total cups sold at Teesta?

वैगई और पेरियार में मिलाकर बेचे गए कुल कॉफी कप, तीस्ता में बेचे गए कुल कप से लगभग कितने प्रतिशत अधिक हैं?
(A) 8%
(B) 10%
(C) 12%
(D) 14%
(E) 16%

Q3: If the same number of tea cups and coffee cups are sold daily at Periyar, then find the total number of cups sold at Periyar in 18 days.
यदि पेरियार में प्रतिदिन चाय कप और कॉफी कप की यही संख्या बेची जाती है, तो 18 दिनों में पेरियार में बेचे गए कुल कपों की संख्या ज्ञात कीजिए।
(A) 6636
(B) 6720
(C) 6840
(D) 6984
(E) 6912

Q4: Find the ratio of total cups sold at Teesta to total cups sold at Mandovi. If the ratio is a:b, find a + b.
तीस्ता में बेचे गए कुल कपों का मांडोवी में बेचे गए कुल कपों से अनुपात ज्ञात कीजिए। यदि अनुपात a:b है, तो a + b ज्ञात कीजिए।
(A) 171
(B) 177
(C) 183
(D) 187
(E) 193

Q5: Tea cups sold at Mandovi are what percent more than the average number of coffee cups sold at Teesta and Vaigai together?
मांडोवी में बेचे गए चाय कप, तीस्ता और वैगई में बेचे गए कॉफी कपों की औसत संख्या से कितने प्रतिशत अधिक हैं?
(A) 30%
(B) 25%
(C) 37 ½%
(D) 40%

(E) 20%

Solutions:

1. Option D
Tea cups at Nayan = $288 \times \frac{5}{4} = 360$
Coffee cups at Nayan = $264 \times \frac{11}{8} = 363$
Total cups at Nayan = 360 + 363 = 723
Coffee cups at Periyar = 234
Difference = 723 − 234 = 489

2. Option C
Coffee cups at Vaigai and Periyar together = 282 + 234 = 516
Total cups at Teesta = 264 + 198 = 462
Difference = 516 − 462 = 54
Required percentage = 54 / 462 × 100 = 11.68% ≈ 12%

3. Option E
Total cups sold daily at Periyar = 150 + 234 = 384
Total cups sold in 18 days = 384 × 18 = 6912

4. Option B
Total cups at Teesta = 264 + 198 = 462
Total cups at Mandovi = 312 + 288 = 600
Required ratio = 462:600 = 77:100
a + b = 77 + 100 = 177

5. Option A
Coffee cups at Teesta = 198
Coffee cups at Vaigai = 282
Average coffee cups = (198 + 282) / 2 = 240
Tea cups at Mandovi = 312
Percentage more = (312 − 240) / 240 × 100 = 30%